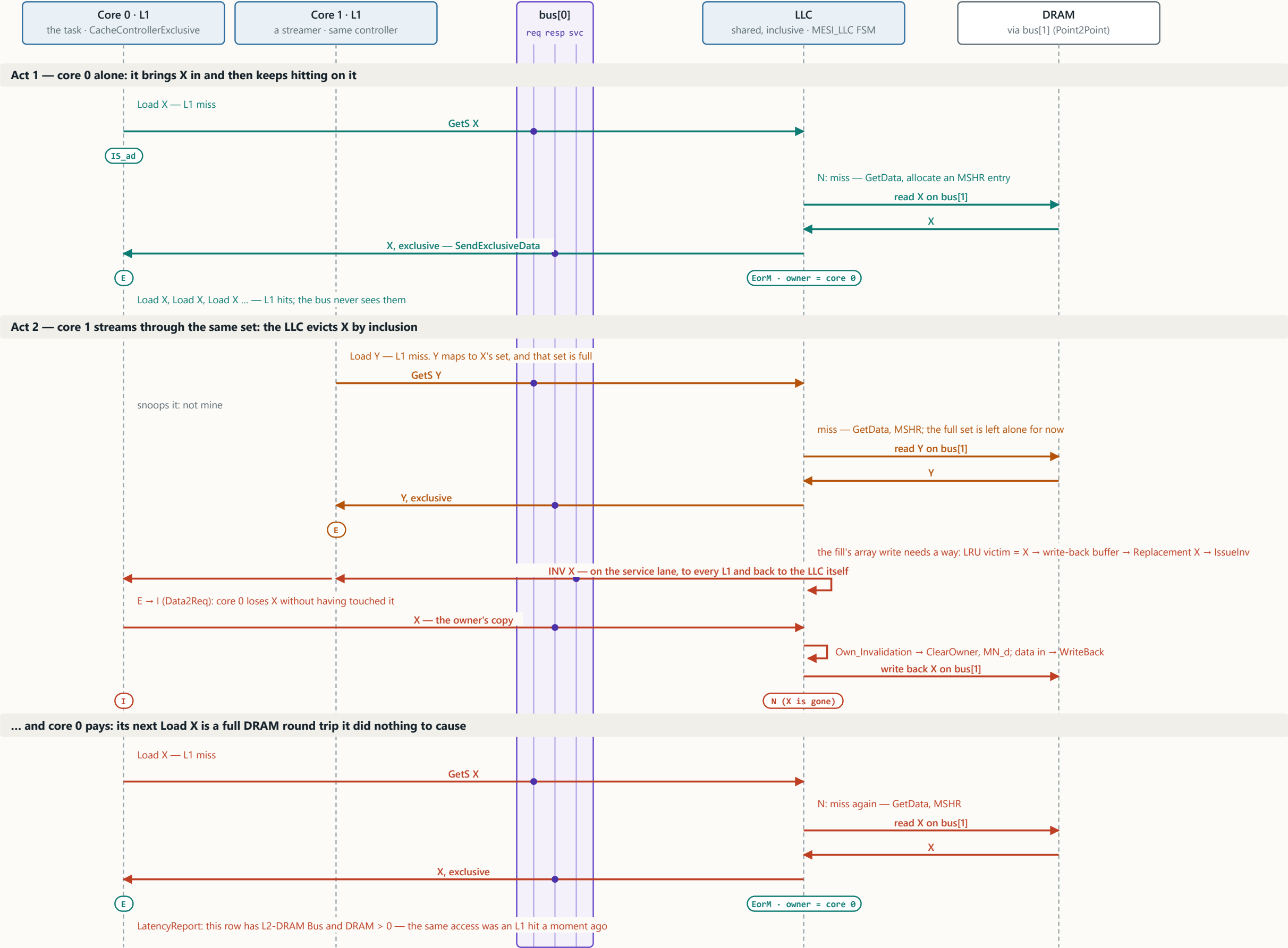


Inclusion interference — two cores, one LLC set, and the line that core 0 loses

Snoop MESI on a TripleBus, shared inclusive LLC (MESI_LLC), time runs downward. Core 0 is the small task of the demo, core 1 one of its streaming neighbours. Teal = core 0's own traffic, amber = core 1's, red = the damage.



In the demo with three streamers this is the steady state of act 2: 53 % of core 0's accesses go to DRAM, none when it runs alone (act 1) — the task's jobs overrun and the robot steers on a stale pose.

What does not help bus arbitration. RR or TDM bounds core 0's wait for a lane, but the INV still arrives and the next access still goes to DRAM: the loss is in the LLC, not on the bus.

What does way reservation (act 3): llc_controller.m_data_handler.way_partition = 0:0 installs core 0's fills in, and evicts from, its own way only. Core 1's Y then takes a way X is not in, and X is never the victim.